SMART ROOM MONITORING SYSTEM

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01 INTRODUCTION

Institutional buildings often waste energy and face safety risks when rooms stay unoccupied. Manual control of lights and fans is frequently neglected, causing unnecessary power use and potential hazards. This project proposes an IoT-based monitoring and automatic power control system using ESP32, which detects human presence through Sensors(FMCW) to reduce energy waste and improve safety

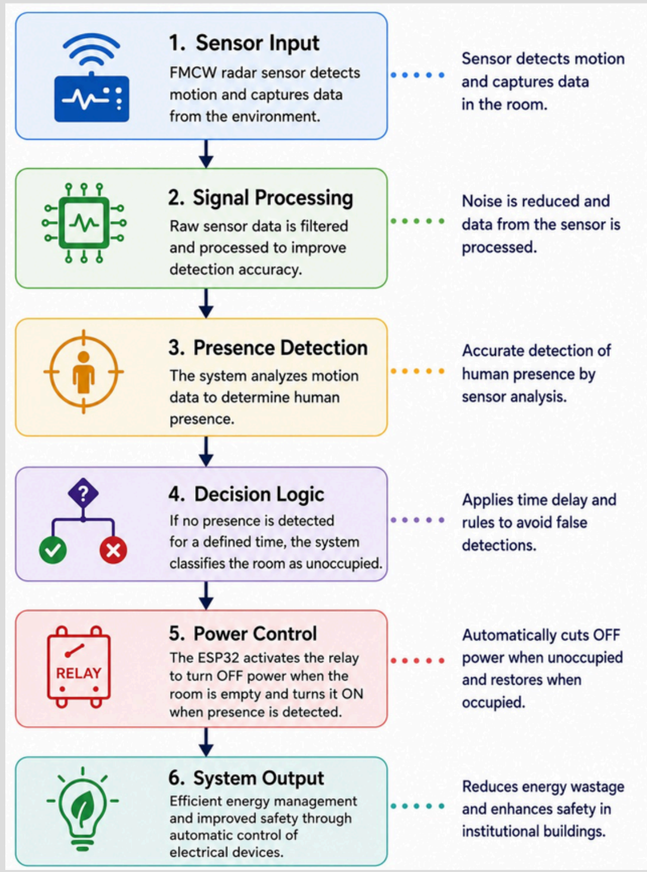
02 OBJECTIVE

- To design and implement an automatic power control system using ESP32
- To detect human presence accurately using Sensors (FMCW)
- To reduce energy consumption through automated control of lights and fans

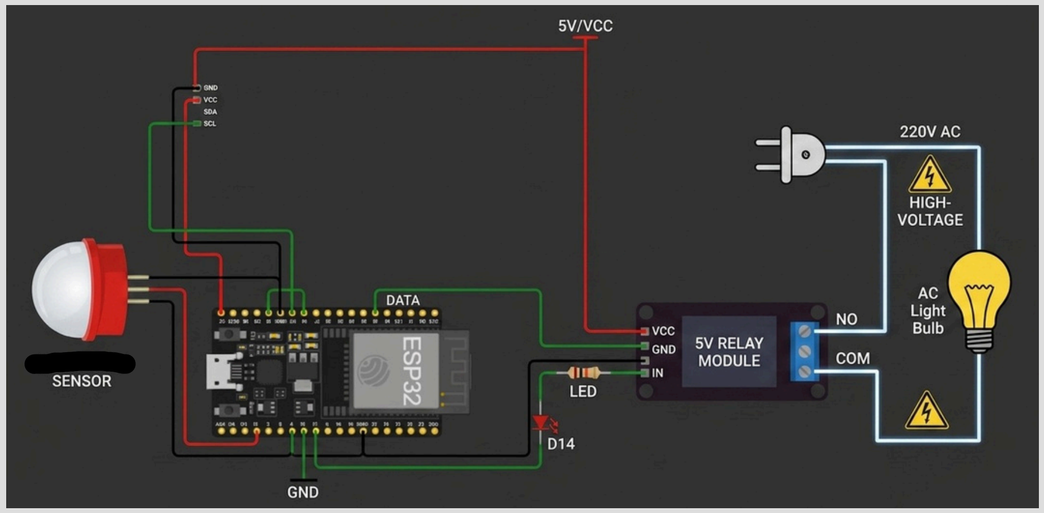
03 EXPECTED OUTCOME

- Efficient automatic switching of electrical devices based on occupancy
- Reduction in energy waste and electricity costs
- Improved safety through continuous monitoring of room usage

04 METHODOLOGY



05 CIRCUIT DIAGRAM



06 CONCLUSION

The IoT-based Monitoring and Automatic Power Control System successfully addresses the critical issues of energy inefficiency and safety in institutional environments. By leveraging the ESP32 and a sensing approach (FMCW), the project demonstrates a highly reliable method for occupancy-driven power management that minimizes false triggers.